

CodSoft Data Science Internship Project Visualizations Report

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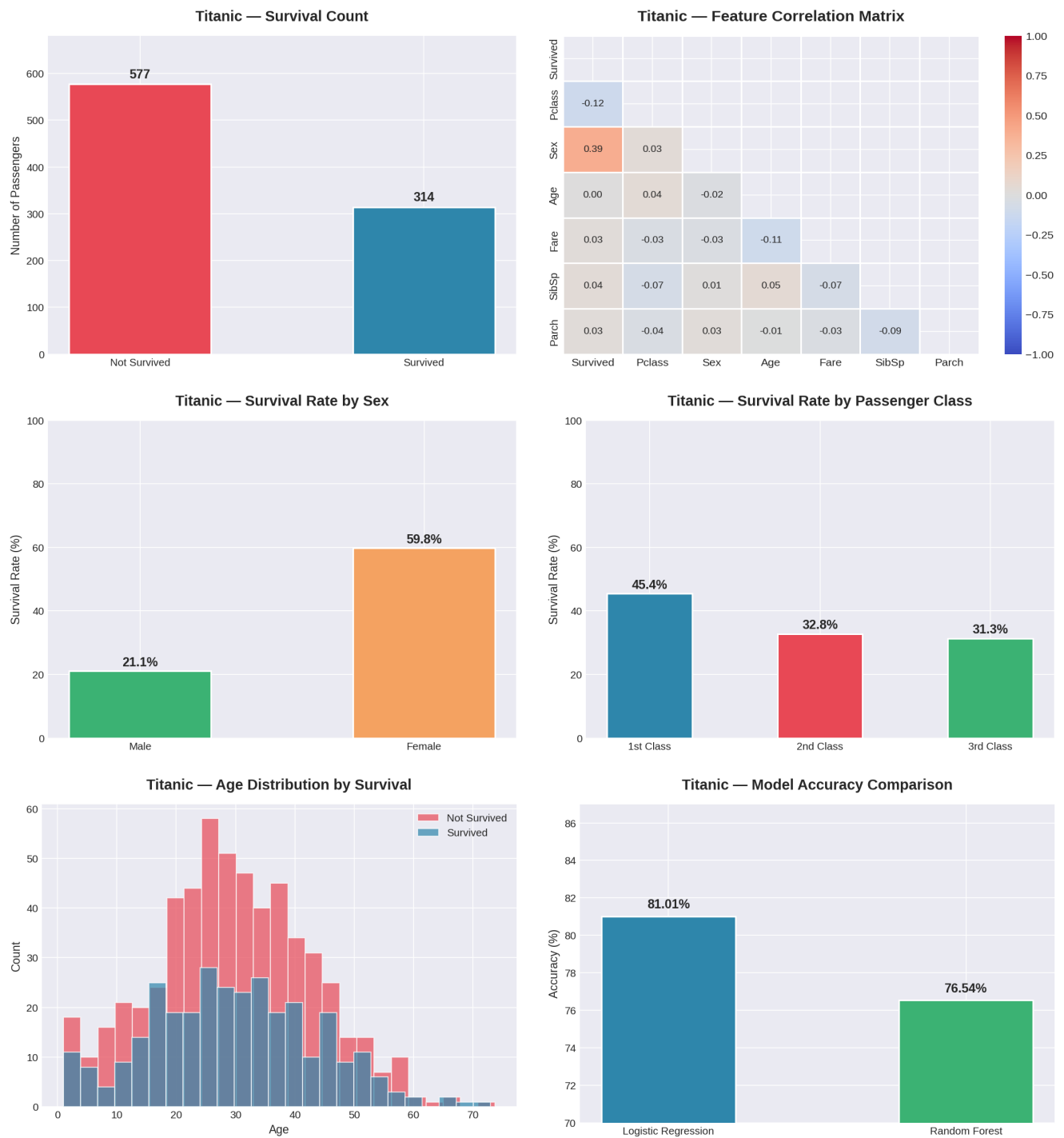
Date: May 06, 2026

#	Project	Best Model	Score
1	Titanic Survival Prediction	Logistic Regression	81.01% Acc
2	Movie Rating Prediction	Random Forest	$R^2=0.1531$
3	Iris Flower Classification	KNN / Random Forest	100% Acc
4	Sales Prediction	Linear Regression	$R^2=0.9003$
5	Credit Card Fraud Detection	Random Forest	F1=0.9987

Project 1: Titanic Survival Prediction

Dataset: 891 passengers | Features: Pclass, Sex, Age, Fare, SibSp, Parch | Target: Survived (0/1)

✓ Best Model: Logistic Regression — Accuracy: 81.01%



Project 2: Movie Rating Prediction

Dataset: 1,000 movies | Features: Genre, Director, Budget, Votes, Year | Target: Rating (1–10)

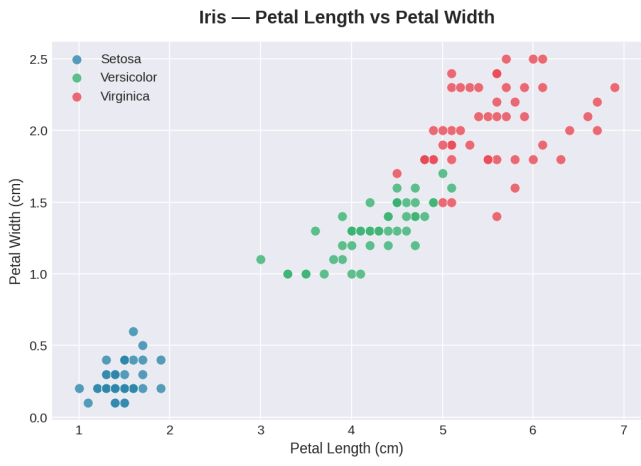
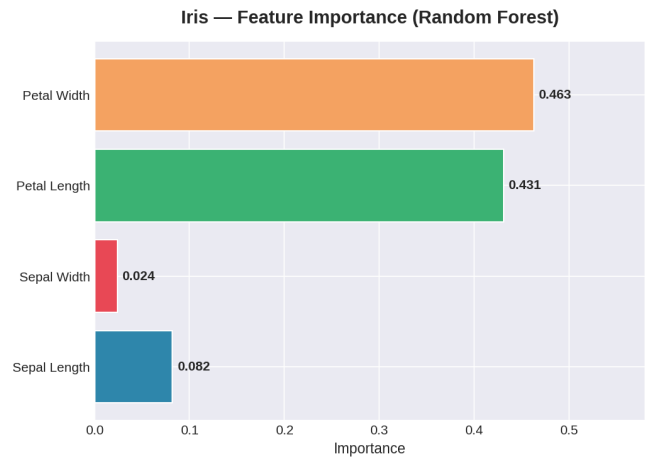
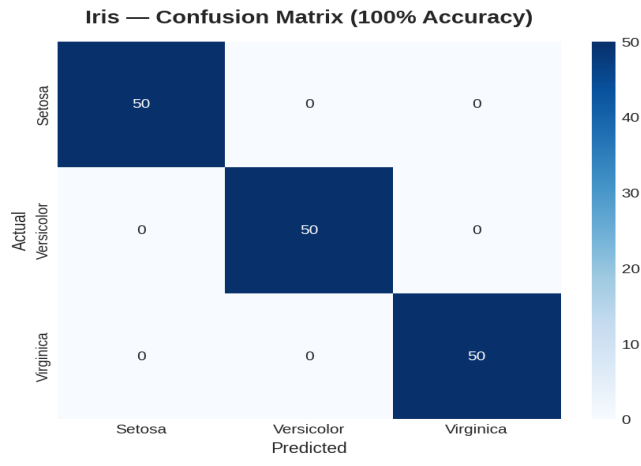
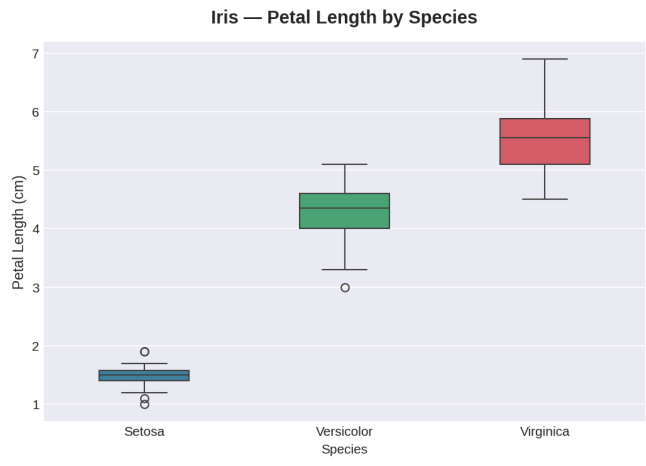
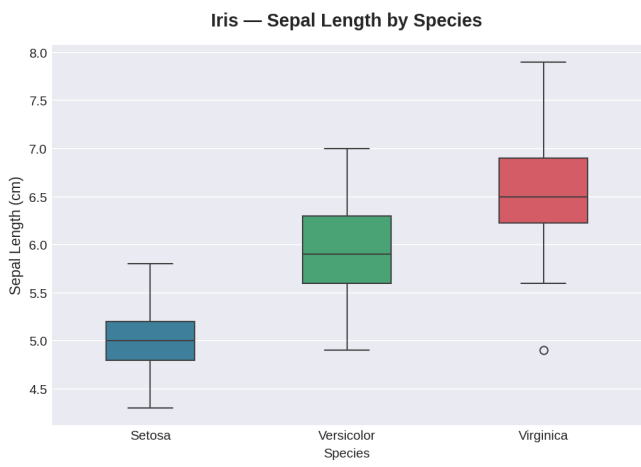
✓ Best Model: Random Forest — $R^2=0.1531$ | $RMSE=0.5283$



Project 3: Iris Flower Classification

Dataset: 150 samples, 3 species (Setosa, Versicolor, Virginica) | Features: 4 measurements (cm)

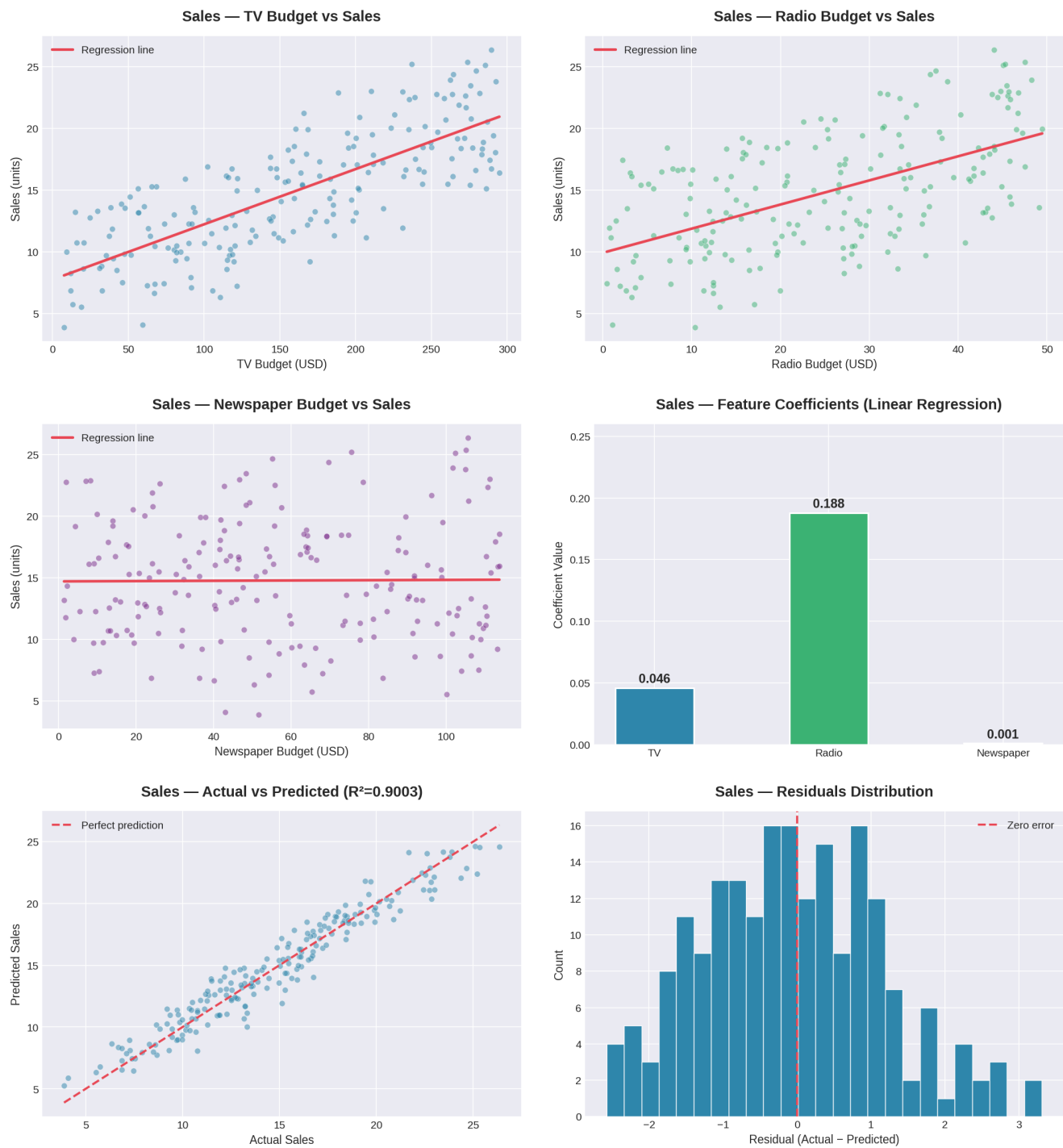
✓ Best Model: KNN / Random Forest — Accuracy: 100%



Project 4: Sales Prediction

Dataset: Advertising (TV, Radio, Newspaper budgets → Sales) | Features: 3 ad channels (USD)

✓ Best Model: Linear Regression — $R^2=0.9003$



Project 5: Credit Card Fraud Detection

Dataset: 2,000 transactions (~2% fraud) | Features: 10 PCA features (V0–V9) + Amount

✓ Best Model: Random Forest — F1=0.9987 | Precision=0.9973 | Recall=1.0 | Acc=99.87%

